Celest Teng

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EDUCATION

Singapore University of Technology and Design — **Singapore**
- Bachelor of Engineering (Computer Science and Design), Honours with Merit — **Sep 2022 to May 2026**
- Minor: Artificial Intelligence, Track: Data Analytics, Visual Analytics and Computing
- Winner of AY2025 Yangzheng Foundation Study Award

WORK EXPERIENCES

ProfilePrint (under SGInnovate's Summation Programme) — **Singapore**
Data Science/ Computer Vision Intern — **May 2025 to Sep 2025**
- Migrated and refactored legacy deep-learning models to PyTorch, enabling modular experimentation and integration of hyperparameter tuning workflows.
- Adapted ProfilePrint's non-visual defect (NVD) modelling pipeline for Honduras using the existing Brazil workflow as reference, extending internal experimentation and evaluation capabilities to a new market.
- Developed and evaluated weakly supervised spectroscopy-based defect detection approaches using Multi-Instance Learning (MIL) and embedding-based representation analysis on sparse industrial datasets with only bag-level supervision.
- Investigated experimental methods for inferring bean-level defect characteristics from spectral signatures and benchmarked alternative modelling approaches to guide future model development.
- Communicated experimental findings, modelling trade-offs, and recommendations to technical stakeholders.

ACADEMIC PROJECTS

Singapore University of Technology and Design — **Singapore**
CC+ Customer Experience Management Platform (Industry Capstone) — **Aug 2025 to May 2026**
- Built and deployed the AI summarisation subsystem of a customer-experience platform delivered to a B2B construction company, transforming customer feedback from WhatsApp, forms, and site visits into structured weekly business insights.
- Evaluated 8 summarisation configurations spanning extractive methods, transformer models, GPT-4o-mini, and Claude, selecting a structured-prompt Claude pipeline augmented with NLP-derived signals for production deployment.
- Owned the AI summarisation track from model evaluation through deployment, designing a provider-agnostic evaluation framework and integrating the workflow into the production platform.

Singapore University of Technology and Design — **Singapore**
Multilingual GPT-2 Evaluation & Interpretability Study, Project Leader — **Aug 2025 to Dec 2025**
- Led multilingual evaluation of GPT-2 across 15 languages, designing experiments to understand how tokenization impacts cross-lingual transfer performance.
- Developed tokenizer fertility metrics and identified token fragmentation as a major bottleneck in multilingual generalization through zero-shot, per-language, and multilingual fine-tuning studies.
- Conducted transformer interpretability analyses including attention entropy, head dispersion, t-SNE representation visualisation, multilingual error taxonomy development, and a linear probing classifier achieving 94% tone-classification accuracy.

Singapore University of Technology and Design — **Singapore**
Structured LLM-Guided Vulnerability Repair for C — **Jan 2026 to May 2026**
- Evaluated structured vulnerability-repair pipelines that combined localisation, classification, and retrieval signals with open-weight LLMs for automated C vulnerability remediation.
- Built evaluation infrastructure for VulnLoc and SAN2VULN benchmarks, including dataset extraction, exploit-verification workflows, pass@k evaluation, and distributed execution tooling.

Singapore University of Technology and Design — **Singapore**
JEPA World Model for Snake Game — **Jan 2026 to May 2026**
- Investigated representation-learning and evaluation pipelines for a JEPA-style world model that learned latent-state dynamics for long-horizon planning tasks.

SKILLS

- Programming: Python, SQL, Git
- Machine Learning: PyTorch, Deep Learning (CNNs, LSTMs, Transformers), Multi-Instance Learning, Weak Supervision, Hyperparameter Tuning, Model Evaluation, Representation Learning
- NLP & LLMs: HuggingFace Transformers, Prompt Engineering, Cross-Lingual Evaluation, Interpretability, Synthetic Dataset Generation
- Data & Analytics: Pandas, NumPy, Scikit-learn, Ray Tune